Energy and Speed Are Related

Chapter 3

Imagine catching a ball in your hand. If the ball is thrown gently, it doesn't hit your hand very hard. But if the same ball is hurled with strength, it hits your hand hard and stings—*OUCH*! Why does it sting? One reason is because of the greater **speed** at which the ball was moving.

Speed is a measure of the distance an object travels during an amount of time. One unit for measuring speed is miles per hour (mph). Many countries use kilometers per hour (kph) as a common unit for speed. When you ride in a car traveling 50 mph (about 80 kph), that means you'll move a distance of 50 miles (or about 80 kilometers) in one hour if the car's speed remains constant.

Big Question

How are energy and speed related?

Vocabulary

speed, n. a measurement of the distance an object travels over an amount of time



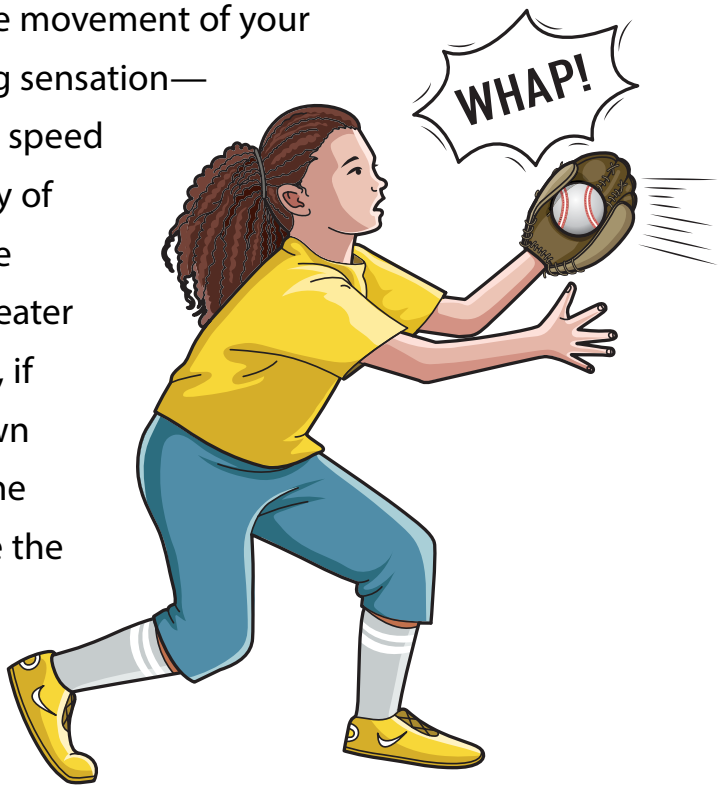
Some pitchers can throw a baseball at speeds faster than 100 miles per hour. Players wear heavy gloves to protect their hands.

Greater Speed Is Related to Greater Energy

Objects with Greater Energy Cause Greater Changes

Remember: energy is the ability to cause change. When you catch a ball, the motion energy from the ball causes a change when it hits your hand. Which will cause greater change—a ball moving slowly or the same ball moving quickly?

The ball moving slowly causes little change—your hand might feel only a gentle tap. Compared to a slow-moving ball, the same ball moving quickly causes greater change when you catch it—your hand moves backward, and you may feel a sting.

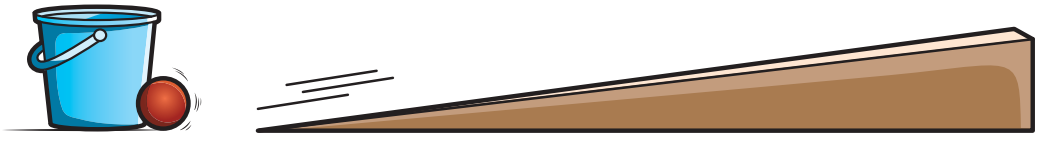


Greater changes—the movement of your hand and the stinging sensation—show that an object's speed is related to its energy of motion. The faster the ball is moving, the greater its motion energy. So, if the same ball is thrown twice, the one with the faster speed will have the greater energy. How do you know? You know because there are greater changes caused by the faster-moving ball when you catch it.

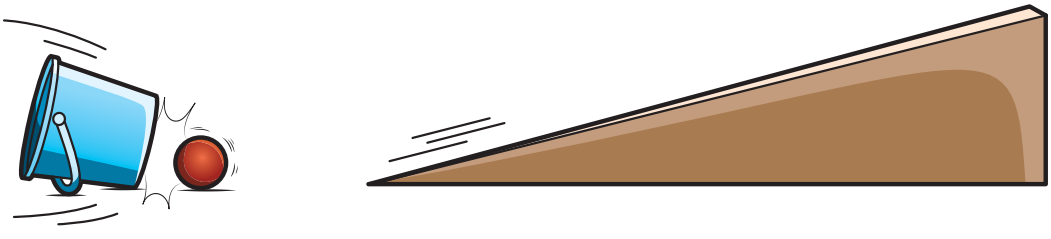
The Faster a Ball Rolls, the More Motion Energy It Has

Objects rolling downhill or down a ramp pick up more speed on steeper slopes. Imagine a ball rolling down a ramp. It will have greater energy of motion after rolling down a steep ramp than it will have rolling down a lower ramp of the same length.

How do we know that a fast-moving ball has more energy than that same ball when it is rolling slowly? If we let the ball hit a bucket at the bottom of the ramp, the slower-moving ball from the lower ramp will not move the bucket very far.



The faster-moving ball from the steeper ramp will knock the bucket over.



The change to the bucket's position is greater when the ball was moving with more speed. That means the motion energy of the ball was greater when the ball was moving faster.

Bouncing a Ball Shows How Energy and Speed Are Related

You can provide evidence to show that an object causes greater change when it is moving quickly compared to the same object when it is moving slowly.

Bounce a ball. First, gently drop the ball. What changes? The ball bounces, but not very high, maybe up to your knees. And it might make a little noise—*boink*.

Then, throw the ball hard against the ground. What changes, compared to when you gently dropped the ball? The ball bounces higher, maybe as high as your head, and it makes a louder noise—*BOING*!

Now think about throwing the ball against the ground with all your strength, as fast as you can. What changes compared to the other times you threw down the ball? It bounces even higher, way above your head, and it makes an even louder noise—*BWANG*!

The motion energy of a fast-moving ball is greater than when the same ball moves slowly. How do you know? You know because you can compare what changes when the ball hits the ground and bounces. When the ball is moving at a faster speed, you know it has greater energy of motion because you can see and hear greater changes. You see a higher bounce and hear a louder sound. Greater change is evidence of greater energy.



A ball bounced softly will result in less change than one bounced hard.